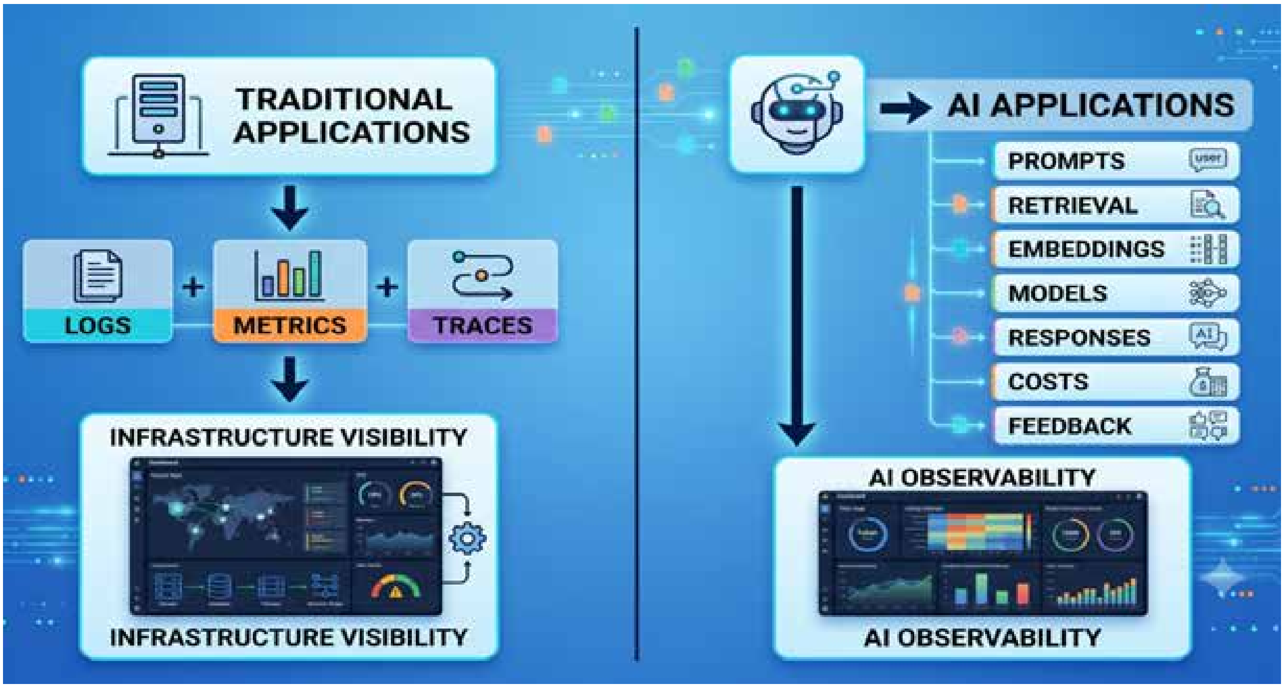

Figure 1: Traditional application monitoring versus AI observability requirements in modern LLM systems

the behaviour of the underlying model. Consequently, engineering teams must track infrastructure performance, model quality, response relevance, retrieval effectiveness, token usage, operational costs, and user satisfaction.

The complexity is further exacerbated by the increasing popularity of retrieval-augmented generation (RAG), AI agents, and multimodal AI systems. Many modern AI applications are made up of several connected pieces such as vector databases, embedding models, orchestration frameworks, prompt templates, external APIs, and foundation models. To identify performance bottlenecks or quality issues across these distributed components, you need a new approach to observability.

This is where AI observability becomes essential. AI observability extends beyond traditional monitoring and provides visibility into the full lifecycle of an AI request -- from user prompt to model response. It enables teams to understand model performance, measure the quality of responses, find failures, monitor costs, and incrementally improve system performance.

Industry trends in AI operations

The last few years have seen a boom in generative AI. The adoption of large language models (LLMs) in customer support systems, enterprise search engines, software development tools, knowledge management solutions, and workflows for business process automation is accelerating.

The advent of RAG architectures has increased operational complexity. Organisations are now combining LLMs with enterprise documents, knowledge bases, and real-time data sources, rather than just model knowledge. This approach improves response accuracy but also introduces new failure modes in document retrieval, embedding quality, indexing, and context relevance.

Another major trend is the rapid increase in AI infrastructure spending. Monitoring inference cost, token consumption, and resource utilisation has become a business-critical requirement when enterprises deploy AI-powered applications at scale. Organisations need to be able to see how models are being used and if the operational costs are sustainable over time.

At the same time, user expectations are increasing. End users expect minimal latency from AI systems, but they also expect the answers to be accurate, relevant, and trustworthy. Even if the infrastructure components are working correctly, poor response quality, hallucinations, or retrieval failures can negatively impact user experience and business outcomes.

These evolving requirements are leading to the emergence of AI operations (AIOps) and AI observability practices. AI observability is emerging as a must-have capability for organisations deploying production-grade AI systems, akin to the way DevOps revolutionised software delivery and SRE improved application reliability.

As a result, engineering teams are increasingly adopting observability platforms that provide visibility into